Supplementary Materials

Supplementary Methods

United States Preventive Service Task Force Criteria for Randomized Controlled Trials and Cohort studies

Criteria 1: Are the initial groups comparable (i.e. randomization, allocation concealment, confounder distribution)?

Criteria 2: Do the groups remain comparable (i.e. attrition, adherence, contamination)?

Criteria 3: Is there a significant differential loss to follow up?

Criteria 4: Were measurement equal, valid and reliable?

Criteria 5: Is the intervention well defined?

Criteria 6: Are the important outcomes considered?

Criteria 7: Is the intention-to-treat analysis used?

Assessment: Good: meets all criteria, Fair: any or all of the problems occur but without fatal flaw, Poor: fatal flaw occurs

Supplementary Table 1. Search Strategy – Example for Search in Embase.com

Breast Cancer terms	Exp breast neoplasms/ OR ((breast OR mamma OR mammary OR lobular) adj4 (Cancer* OR carcinoma* OR neoplas* OR tumor* OR tumour* OR malignan*).mp.) 'breast cancer'/exp OR ((breast OR mamma OR mammary OR lobular) near/4 (Cancer* OR carcinoma* OR neoplas* OR tumor* OR tumour* OR malignan*)):ti,ab,kw,de
Menopause Status terms	AND (exp Menopause/ OR (menopaus* OR premenopaus*).ti,ab,hw,kf. OR (menopauz* OR pre-menopaus* OR perimenopaus* OR climacteric OR climacterium).ti,ab. OR (change of life).ti,ab. OR ((pre OR post) ADJ1 (menopaus*)).ti,ab.) ('menopause'/exp OR 'premenopause'/exp OR (menopaus* OR premenopaus*):ti,ab,de OR (menopauz* OR pre-menopaus* OR perimenopaus* OR climacteric OR climacterium OR 'change of life'):ti,ab OR ((pre OR post) NEAR/1 (menopaus*)):ti,ab)
Mammographic Breast Density terms	(exp Breast Density/ OR (volumetric breast).ti,ab. OR ((mammogram* OR mammographic*) ADJ3 (density OR densities OR dense)).ti,ab. OR ((breast OR mammary) ADJ2 (density OR densities)).ti,ab.) ('breast density'/exp OR 'volumetric breast' OR ((mammogram* OR mammographic*) NEAR/3 (density OR densities OR dense)):ti,ab OR ((breast OR mammary) NEAR/2 (density OR densities)):ti,ab)

Supplementary Table 2. Inclusion and Exclusion Criteria for Study Selection

	Inclusion	Exclusion
Population	Premenopausal women with no prior history of invasive breast cancer. Ductal carcinoma in situ, lobular carcinoma in situ and atypical hyperplasia were included. We have allowed for the inclusion of high-risk women, as most studies used this as inclusion criteria for their study populations Premenopausal status was deferred to the author's choice of premenopausal definition.	Postmenopausal women. Women with prior history of invasive breast cancer.
Intervention	Any pharmacological agent hypothesized to reduce MBD, reported by dose and duration of intervention	Hormone replacement studies as it was assumed that population was peri- or post-menopausal. Dietary (e.g. milk) or exercise interventions. Non-specific agents (e.g. multivitamins or supplements not otherwise defined) or without a reported specific dosage.
Comparison	Control/comparison group required	
Outcome	Effect on mammographic density quantitatively reported for premenopausal women. Secondary outcomes include development of breast cancer and adverse events reported.	
Time frame	Any publication through the date of literature search on January, 2020	
Study design	Published interventional study	Observational studies, commentaries, editorials, reviews, abstracts, meta-analysis, systematic reviews
Setting	Any country/setting	
Language	English, Spanish, German, French	Studies published in different languages than the included

Supplementary Table 3. Summary of quality assessment per United States Preventive Service Task Force Criteria for Randomized Controlled Trials and Cohort studies

Article	Criteria 1	Criteria 2	Criteria 3	Criteria 4	Criteria 5	Criteria 6	Criteria 7	Summary score
Brisson, et al., 2017 (26)	Yes	Yes	No	Yes	Yes	Yes	Yes	Good
Pasta, et al., 2015 (27)	Yes	Yes	No	No	Yes	Yes	No	Poor
Powles, et al., 2008 (28)	Yes	Yes	No	Yes	Yes	Yes	Not reported	Fair
Cuzick, et al., 2004 (15)	Yes	Yes[a]	No[a]	Yes	Yes	Yes	Yes [a]	Fair
Maskarinec, et al., 2003 (25)	Yes	Yes	No	Yes	Yes	Yes	Not reported	Good
Howell, et al., 2018 (30)	Yes	Yes	No	Yes	Yes	Yes	Yes	Good
Gram, et al., 2001 (29)	Not reported	Not reported	No	Yes	Yes	Yes	No	Poor

[a] inclusion criteria required complete adherence to intervention